

LiteTest

LiteTest is a lightweight framework and Application Programming Interface (API) for defining, running, and reporting tests in C/C++ projects. It provides a simple core macro-driven Runner API plus a function-based Test API, fault-tolerant execution, and clear reporting. It is ideal for small to medium C projects that need reliable testing without heavy tooling and dependencies. It can be used for unit and command-line testing.

LiteTest provides a Runner API and a Test API, each implemented with a single .h / .c pair with no external dependencies and requiring only a POSIX.1-2001 environment and a C99-compliant compiler.

Copyright (c) 2026 Paul Sinclair

License

SPDX-License-Identifier: MIT

Permission is hereby granted, free of charge, to any person obtaining a copy of this software and associated documentation files (the “Software”), to deal in the Software without restriction, including without limitation the rights to use, copy, modify, merge, publish, distribute, sublicense, and/or sell copies of the Software, and to permit persons to whom the Software is furnished to do so, subject to the following conditions:

The above copyright notice and this permission notice shall be included in all copies or substantial portions of the Software.

THE SOFTWARE IS PROVIDED “AS IS”, WITHOUT WARRANTY OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO THE WARRANTIES OF MERCHANTABILITY, FITNESS FOR A PARTICULAR PURPOSE AND NONINFRINGEMENT. IN NO EVENT SHALL THE AUTHORS OR COPYRIGHT HOLDERS BE LIABLE FOR ANY CLAIM, DAMAGES OR OTHER LIABILITY, WHETHER IN AN ACTION OF CONTRACT, TORT OR OTHERWISE, ARISING FROM, OUT OF OR IN CONNECTION WITH THE SOFTWARE OR THE USE OR OTHER DEALINGS IN THE SOFTWARE.

Preface

For LiteTest documentation and the repository layout, see the **LiteTest Documentation Guide**.

Table of Contents

1. Introduction	1
1.1. Key Features	1
1.2. Quick Start	1
1.3. Requirements	2
1.4. Installation	3
2. LiteTest Runner Framework and API	4
2.1. What the Runner Framework and API Does	4
3. Running Tests	5
4. Report Generation	6
5. Customization	7
6. Test Report	8
7. Advanced Features and Topics	9
8. Test Expressions	10
9. Setup and Teardown	11
10. Test Groups	12
11. Concurrent Tests	13
12. Isolation	14
13. LiteTest Test API	15
13.1 File Functions	15
13.2 Compare Functions	15

1. Introduction

1.1. Key Features

- Pure C implementation
- Minimal footprint (single header + source)
- Fault detection using POSIX signals
- Optional process-isolated execution
- Parallel and concurrent test execution
- Clear pass/fail/fault reporting
- Simple API based on macros and test group functions

1.2. Quick Start

A test executable consists of your tests as C/C++ expressions/functions, and the orchestrator and test group functions you write using the LiteTest API that manages the execution.

1. Copy 'litetest_runner.h' and 'litetest_runner.c' to your current directory:

```
cp /path/to/litetest_runner.h .
cp /path/to/litetest_runner.c .
```

2. Create a file named `test_quick.c` in the same directory.

3. Copy and paste the code into `test_quick.c`:

```
#include "litetest_runner.h"

// A simple test group function.

static LT_DECLARE_GROUP(test_quick)
{
    LT_INIT_GROUP(test_quick, 1);

    int a = 2;
    int b = 2;

    // 4 test assertions.
    LT_TEST(a == b, , 0);           // Pass
    LT_TEST(a + b == 4, , 0);       // Pass
    LT_TEST(a - b == 1, , 0);       // Fail
    LT_TEST(LT_FAULT(1), , 0);      // Fault

    LT_RETURN;
}

// A simple orchestrator (main) function.

LT_DECLARE_ORCHESTRATOR(main)
{
    LT_INIT_ORCHESTRATOR(main, quick, 1);
```

```

LT_PARSE_ARGS(2, "quick_test_report.txt");
LT_OPEN_REPORT("Test Quick Report");

// Single test category.
LT_WRITE_RESULT(LT_GROUP(test_quick), "Quick tests");

LT_CLOSE_REPORT("Note: This report is a very simple example of using LiteTest.\n"
    "Note: Multiple test categories can be added using multiple\n"
    "    test functions.\n"
    "Note: Orchestrator (`main`) and test functions can be placed in\n"
    "    individual modules (.c files).\n"
    "Note: Parameters can be set to run tests in parallel, isolate\n"
    "    a test to a separate thread or process, etc.\n"
    "Note: The expression for LT_TEST can reference functions to\n"
    "    provide a more complex test. A non-zero result indicates\n"
    "    pass and a zero result indicates fail. If a fault occurs\n"
    "    executing the expression, it is detected and counted in\n"
    "    the report as a fault.\n"
    "Note: Larger projects can place files in a more conventional\n"
    "    layout (e.g., `include/` and `src/`, but this example keeps\n"
    "    everything in your current directory for simplification.\n"
    "Note: See README.md for LiteTest for additional API features.\n");
LT_EXIT;
}

```

4. Build the executable `test_quick` in your current directory:

```
cc -std=c99 -Wall -Wextra -o test_quick test_quick.c litetest_runner.c
```

5. Run it:

```
./test_quick
```

6. View `quick_test_report.txt` in your current directory:

```
less quick_test_report.txt    # Press 'q' to quit
```

Example of the report:

```

LiteTest Report

```

	Pass	Fail	Fault
1. Orchestrator	4		
2. Guard 1 and 2	6		
Total	10		

1.3. Requirements

Supported compilers, C standard level, and any platform notes.

1.4. Installation

How to add LiteTest to your project, including copying the header, adding the source file, or integrating via your build system.

2. LiteTest Runner Framework and API

The Runner framework (implemented using the Runner API) executes all test expressions (that implement the tests), aggregates results, and produces the report. This is the framework used by your test executable.

The Runner API provides typedefs, enums, macros, and functions for writing tests. Developers use this API to express expected behavior, group related tests, and define optional setup/teardown logic.

2.1. What the Runner Framework and API Does

TODO

3. Running Tests

How to invoke the runner from `main()`, including optional parameters such as output paths or configuration flags.

4. Report Generation

How LiteTest collects pass/fail information and formats it for output.

5. Customization

Notes that you can build your own runner logic if you need custom ordering, filtering, or integration behavior.

6. Test Report

Describes the structure of the test report, failures, optional details, and how to consume it in tooling or CI.

7. Advanced Features and Topics

Filtering tests and custom report writers are available in the User Guide.

8. Test Expressions

How to write a test expression for `LT_TEST(expr)` and how expressions are evaluated.

9. Setup and Teardown

Optional per-group or per-test initialization and cleanup helpers.

10. Test Groups

How to organize related tests into groups for readability and logical structure.

11. Concurrent Tests

How to mark tests as concurrent and how LiteTest handles parallel execution safely.

12. Isolation

isolation ...

13. LiteTest Test API

The Test API provides macros and functions for writing tests. Developers use this API to express expected behavior, group related tests, and define optional setup/teardown logic.

13.1 File Functions

Examples of file functions:

13.2 Compare Functions

Examples of compare functions: